

ARK-448 Results — Dynamical Decoupling vs. Baseline

Outcome: ABORTED AT SPAM GATE (honest gate-stop).
No DD-vs-baseline data was obtained. No verdict is claimed.

Summary

ARK-448 was locked, the qubit pair was selected, and the preregistered SPAM gate was executed on real IBM hardware (`ibm_marrakesh`). **The SPAM gate FAILED**, so — per the LOCKED preregistration — the principal 4-arm job was **not** submitted and execution was halted. This is a preregistered abort, not a result about dynamical decoupling.

What ran

Step	Status
LOCK (preregistration + code, tag <code>ark-448-v1.0-lock</code>)	✓ done
Qubit selection (Q_A=1, Q_P=2, <code>ibm_marrakesh</code>)	✓ done
SPAM gate (job <code>d9cqji4inv1c73ao54eg</code> , 2048 shots)	✗ FAILED
Principal 4-arm job (8192 shots)	⊖ NOT RUN (halted per protocol)

SPAM gate detail

Check	Measured	Threshold	Pass?
SPAM_A (authorizer 1) readout error)	0.01025	≤ 0.02	✓
SPAM_P (payload +) symmetry, P(1)−0.5)	0.02197	≤ 0.02	✗

The payload readout symmetry deviated **0.02197** from the ideal 0.5, exceeding the preregistered ± 0.02 tolerance by **0.00197**. At 2048 shots the binomial standard deviation at $p=0.5$ is ≈ 0.011 , so the observed deviation is $\approx 2.0\sigma$ — consistent with a marginal statistical fluctuation and/or minor readout drift on the calibration snapshot used.

Why we did not re-run or loosen the gate

The ExecutionProof protocol is **preregistration-first**. The LOCKED preregistration (ARK_448_preregistration.md , Section 5) fixed the SPAM_P tolerance at ± 0.02 before execution and does **not** permit:

- re-running the gate until it passes (that would be selection bias / p-hacking on the gate),
- loosening the threshold post-hoc, or
- proceeding to the principal job on a failed gate.

Honoring the lock, we recorded the failure and stopped. This mirrors the honest FAIL in ARK-445: the value of the protocol is precisely that it halts on marginal conditions instead of manufacturing a clean-looking result.

What this does and does not mean

- It does **NOT** mean dynamical decoupling failed, succeeded, or was measured at all — the DD-vs-baseline comparison never executed.
- It **does** mean the readout condition on this snapshot/qubit fell just outside the preregistered tolerance, and the gate did its job.
- Budget was conserved: the expensive principal job was never submitted.

Reproducing / re-attempting

A future ARK-448 attempt would re-lock (new tag), re-select qubits against a fresh calibration snapshot, and re-run the SPAM gate. Because the gate is preregistered, a re-attempt is a **new locked run**, not a silent retry of this one. This run stands in the record as a genuine gate-stop.

Files

- proofrecord.json — machine-readable gate-stop record (real SPAM data, no principal data).
- spam_results.json — raw SPAM gate counts and computed errors.
- selected_qubits.json — selected pair and readout errors.
- calibration_snapshot_ibm_marrakesh_20260717.json — backend calibration at submission.
- RUN_LOG.md — chronological execution log.